MILITARY SPECIFICATION

TIMER, INTERVAL DUMMY, XM10 AND POWER SUPPLY, DUMMY PS406D ASSEMBLY

1. SCOPE

1.1 This specification covers special requirements not included on the drawings, and provides the quality assurance provisions for the fabrication of parts, assembling, and the preparation for delivery for one type of interval timer, and power supply assembly for training troops and designated as Interval Timer, Dummy, XM10 and Power Supply, Dummy PS406D Assembly. This specification is limited to repair part procurement.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this SQAP to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-L-4343 - Lubricating Grease.
MIL-P-116 - Preservation, Methods of.
MIL-S-6872 - Soldering Process, General specification for.

STANDARDS

MILITARY

BFDX3 - Pin, Steel, Straight.
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168 - Lot Numbering of Ammunition.
MS9048 - Pin Spring.

DRAWINGS

ORDNANCE CORPS

8794342 - Seal, Metallic, Assembly, Details and Instructions for Application.
8806652 - Marking and Sealing for Cylinderical Containers.
8808362 - Contr, Pkg for Conus Shipments.

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8823675 - Interval Timer, Dummy, XM10 and Pwr. Supply Dummy PS406D Assembly.
8823710 - Timer, Interval, Dummy, XM10 Assembly.
8823711 - Case.
8823712 - Knob.
8858967 - Drum, Metal, Shipping, for Timer, Interval and Power Supply w/Battery, Thermal.
8859013 - Drum, Metal, Shipping (Exterior).
10965479 - Power Supply Assembly PS406D.
10965487- Ring, Conductor No. 1.
10965488- Ring, Conductor No. 2.
10965489- Ring, Conductor No. 3.
10965490- Ring, Conductor No. 4.
10965491- Conductor Ring Assembly.
10966863- Case, Blank.
10966864- Case, w/Primer Holder.
10966866- Holder, Primer.
10966909- Case, Assembly.
10966910- Power Supply Assembly PS406D without Conductor Ring.
10966913- Conductor Body.
10974210- Resistor, Fixed, Wire Wound (Power Type).
10974211- Case Marking PS406D
ET-8823675 - Equipment Tabulation

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material used to fabricate any part complies with the specifications listed on the applicable drawing.

3.2 Assembly. The assembly shall comply with all requirements specified on Drawing (dwg.) 8823675, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawing and herein.

3.3 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.4 Circuit resistance (see dwg. 10965479). - The circuit resistance of any circuit shall comply with table I when tested as specified in 4.4.1.

Table I

Pin	Pin	Ohms
1	2	Infinity
1	3	1.2 ± 5%
1	4	1.2 ± 5%

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3.5 Insulation resistance (see dwg 10965479). - The insulation resistance shall not be less than 20 megohms when tested as specified in 4.4.2.

3.6 Workmanship.

3.6.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.6.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.6.3 Burrs. Parts shall be free from burrs.

3.6.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.6.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.6.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may use his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection plan. The contractor is responsible for compliance with all dimensions and requirements specified on applicable drawings and herein. The contractor's approved inspection plan shall include, as a minimum, all the inspections specified herein.

4.2 First article inspection. Not applicable.

4.3 Product acceptance inspection.

4.3.1 Lot formation. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristic in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Major characteristic table II		Minor characteristic table III	
Lot size	Sample size	Lot size	Sample size
1- 25	100% inspection	1- 25	100% inspection
26- 30	25		
31- 35	26	26- 35	16
36- 40	27		
41- 45	28	36- 45	17
46- 50	29		
51- 55	30	46- 55	18
56- 60	31		
61- 65	32	56- 65	19
66- 70	33		
71- 75	34	66- 100	20
76- 85	35		
86- 95	36	101- 200	21
96- 110	37		
111- 130	38	201 and over	22
131- 156	39		
157- 190	40		
191- 240	41		
241- 340	42		
341- 520	43		
521-1120	44		
1121 and over	45		

4.3.2.1 Classification of characteristics. Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Case, blank (see dwg 10966863 covering a detail of dwg 10966864).

Categories	Characteristics	Method of inspection
Critical:	None defined.	
Major:		
101.	Outside diameter	Gage
102.	Distance from bottom of large inside diameter to bottom of case	Gage
103.	Location of bottom hole	Gage
104.	Perpendicularity of bottom	Gage
105.	Material (see 3.1)	Visual
Minor:		
201	Workmanship (see 3.6)	Visual/Manual

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4.3.2.1.2 Holder, primer (see dwg 10966866 covering a detail of dwg 10966864).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Total height.	.Gage
102.	Distance from top surface to shoulder	.Gage
103.	Pilot diameter.	.Gage
104.	Pitch diameter of thread.	.Gage
105.	Minor diameter, max	.Gage
106.	Thickness to pilot diameter	.Gage
107.	Material (see 3.1).	.Visual
Minor:		
201.	Workmanship (see 3.6)	.Visual/Manual

4.3.2.1.3 Case w/primer holder (see dwg 10966864 covering a detail of dwg 10966909).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Soldering	.Visual
102.	Objective evidence (see 6.1) available for electroplate in accordance with dwg 10966864.	.Visual
Minor:		
201.	Workmanship (see 3.6)	.Visual/Manual

4.3.2.1.4 Case, assembly (see 10966909 covering a detail of dwg 10966910).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Proper primer assembly	.Visual/Manual
Minor:		
201.	Workmanship (see 3.6)	Visual/Manual

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4.3.2.1.5 Power supply assy P5406D without conductor ring (see dwg 10966910 covering a detail of dwg 10965479).

Categories	Characteristics	Method of inspection
Critical:	None defined.	
Major:		
101.	Objective evidence (see 6.2) available for resistors in accordance with dwg 10974210.	Visual
102.	Objective evidence (see 6.2) available for paint in accordance with dwg 10966910.	Visual
103.	Soldering in accordance with MIL-S-6872.	Visual/Manual
104.	No paint on header, pins and thread.	Visual
105.	Weight	Gage
Minor:		
201.	Workmanship (see 3.6).	Visual/Manual

4.3.2.1.6 Conductor body (see dwg 10966913 covering a detail of dwg 10965491).

Categories	Characteristics	Method of inspection
Critical:	None defined.	
Major:		
101.	Height of center projection, max.	Gage
102.	Depth of cavities	Gage
103.	Total thickness	Gage
104.	Outside taper present	Visual
105.	Outside diameter.	Gage
106.	Large diameter of outer cavity, max	Gage
107.	Small diameter of outer cavity, min	Gage
108.	Large diameter of inner cavity, max	Gage
109.	Diameter of center projection, min.	Gage
110.	Material (see 3.1)	Visual
Minor:		
201.	Workmanship (see 3.6).	Visual/Manual

4.3.2.1.7 Ring, conductor No. 1 (see dwg 10965487) and Ring conductor No. 3 (see dwg 10965489) covering a detail of dwg 10965491).

Categories	Characteristics	Method of inspection
Critical:	None defined.	

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Categories	Characteristics	Method of inspection
Major:		
101.	Total thickness	Gage
102.	Chamfer present	Visual
103.	Inside diameter	Gage
104.	Outside diameter, min	Gage
105.	Concentricity of outside diameter	Gage
106.	Material (see 3.1)	Visual
107.	Objective evidence (see 6.2) available for protective finish in accordance with dwg 10965487	Visual
Minor:		
201.	Workmanship (see 3.6)	Visual/Manual

4.3.2.1.8 Ring, conductor No. 2 (see dwg 10965488 covering a detail of dwg 10965491).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Outside diameter	Gage
102.	Outside taper present	Visual
103.	Inside diameter, max	Gage
104.	Thickness	Gage
105.	Concentricity of outside diameter	Gage
106.	Material (see 3.1)	Visual
107.	Objective evidence (see 6.2) available for protective finish in accordance with dwg 10965488	Visual
Minor:		
201.	Workmanship (see 3.6)	Visual/Manual

4.3.2.1.9 Ring, conductor No. 4 (see dwg 10965490 covering a detail of dwg 10965491).

Categories	Characteristics	Method of inspection
Critical:		
Major:		
101.	Total length	Gage
102.	Depth of cavity	Gage
103.	Diameter of cavity, max.	Gage
104.	Concentricity of diameter of cavity.	Gage
105.	Outside taper present.	Visual
106.	Outside diameter	Gage
107.	Material (see 3.1)	Visual
108.	Objective evidence (see 6.2) available for protective finish in accordance with dwg 10965490.	Visual/Manual

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Categories	Characteristics	Method of inspection
Minor:		
	201. Workmanship (see 3.6)	Visual/Manual

4.3.2.1.10 Conductor ring assembly (see dwg 10965491 covering a detail of dwg 10965479).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
	101. Conductor rings tight	Manual
	102. No excess epoxy	Visual
Minor:		
	201. Workmanship (see 3.6)	Visual/Manual

4.3.2.1.11 Power supply, dummy PS406D (see dwg 10965479 covering a detail of dwg 8823675).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
	101. Total length	Gage
	102. Depth of thread cavity	Gage
	103. Depth of pilot diameter	Gage
	104. Perpendicularity of right end	Gage
	105. Outside diameter	Gage
	106. Concentricity of outside diameter	Gage
	107. Inside pilot diameter	Gage
	108. Pitch diameter of thread	Gage
	109. Minor diameter of thread, max.	Gage
	110. Concentricity of thread	Gage
	111. No excess solder	Visual/Manual
	112. No excess epoxy	Visual
	113. Grease, oil or other similar material not present	Visual/Manual
Minor:		
	201. Serialization (see 3.5)	Visual
	202. Marking (see dwg 10974211)	Visual
	203. Workmanship (see 3.6)	Visual/Manual

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4.3.2.1.12 Pin, steel, straight (see dwg BFDX3 - 505254) and Pin-spring (see dwg MS9048-101) covering details of dwg 8823710).

Categories	Characteristics	Method of inspection
Critical:	None defined.	
Major:		
101.	Objective evidence (see 6.2) available for part in accordance with applicable standard.	Visual
Minor:		
201.	Workmanship (see 3.6)	Visual/Manual

4.3.2.1.13 Case (see dwg 8823711 covering a detail of dwg 8823710).

Categories	Characteristics	Method of inspection
Critical:	None defined.	
Major:		
101.	Location and contour of opening	Visual
102.	Large diameter of groove in large flange.	Gage
103.	Small diameter of groove in large flange.	Gage
104.	Diameter of knob spindle.	Gage
105.	Location of hole in knob spindle.	Gage
106.	Diameter of hole in knob spindle.	Gage
107.	Diameter of projection right of large flange.	Gage
108.	Depth of small right cavity	Gage
109.	Depth of pin hole small right cavity.	Gage
110.	Pitch diameter of thread.	Gage
111.	Major diameter of thread, min	Gage
112.	Diameter of pin hole in small right cavity.	Gage
113.	Small outside diameter right end.	Gage
114.	Outside body diameter	Gage
115.	Width of thread clearance, max.	Gage
116.	Distance from end of thread to shoulder	Gage
117.	Length of threaded projection	Gage
118.	Length of body.	Gage
119.	Depth of groove in large flange	Gage
120.	Material (see 3.1).	Visual
121.	Objective evidence (see 6.2) available for protective finish in accordance with dwg 8823711. . .	Visual
122.	Depth of contoured recess	Gage
Minor:		
201.	Marking	Visual
202.	Workmanship (see 3.6)	Visual/Manual

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4.3.2.1.14 Knob (see 8823712 covering a detail of dwg 8823710).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Diameter of shaft hole	Gage
102.	Location of pin hole.	Gage
103.	Diameter of pin hole	Gage
104.	Material (see 3.1)	Visual
105.	Objective evidence (see 6.2) available for protective finish in accordance with dwg 8823712	Visual
Minor:		
201.	Marking.	Visual
202.	Workmanship (see 3.6).	Visual/Manual

4.3.2.1.15 Timer, interval, dummy, XM10 assembly (see dwg 8823710 covering a detail of dwg 8823675).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Dial readable thru window.	Visual
102.	No part missing or incorrectly assembled	Visual
Minor:		
201.	Serialization (see 3.5).	Visual
202.	Stamping	Visual
203.	Workmanship (see 3.6).	Visual/Manual

4.3.2.1.16 Interval timer, dummy, XM10 and power supply (see dwg 8823675).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Objective evidence (see 6.2) available for lubricating grease in accordance with MIL-L-4343.	Visual
102.	Grease on gasket present	Visual
103.	Properly assembled	Visual
Minor:		
201.	Workmanship (see 3.6)	Visual/Manual

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4.3.2.1.17 Drum, metal, shipping, for timer, interval and power supply w/battery, thermal (see dwg 8858967 and section 5).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Objective evidence (see 6.2) available for quick leak test on barrier bag after packing and sealing in accordance with MIL-P-116.	.Manual
102.	Objective evidence (see 6.2) available for Drum in accordance with dwg 8859013, part No. 8859013-16.	.Visual
103.	Contents tight in container	.Manual
104.	Material (see 3.1).	.Visual
105.	Objective evidence (see 6.2) available for sealing cover in accordance with dwg 8806652.	.Visual
106.	Objective evidence (see 6.2) available for seal in accordance with dwg 8794342	.Visual
Minor:		
201.	Workmanship (see 3.6)	.Visual/Manual

4.3.2.1.18 Container, packing, for conus shipments (see dwg 8808362).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major: None defined.		
Minor:		
201.	Material (see 3.1).	.Visual
202.	Marking	.Visual
203.	Item properly packed.	.Visual

4.3.3 Testing. The contractor shall perform all the tests specified in this paragraph. Tests do not have to be performed in the order specified.

	Test	Test Classification
a.	Circuit resistance	Major
b.	Insulation resistance	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

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4.3.3.1 Circuit resistance (see 3.4) and insulation resistance (see 3.5). The circuit resistance and insulation resistance shall be tested on each power supply assembly as specified in 4.4.1 or 4.4.2 as applicable. If any power supply assembly does not comply with 3.4 and 3.5, the power supply assembly shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions and calibration instructions specified on the Equipment Lists EL. EL's for complete assemblies are listed on the Equipment Tabulation (ET-3823675). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment, and obtain approval of such designs together with any necessary operating instructions and special calibration instructions, from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4:

a. Any deviation from equipment or procedures specified on the EL is desired.
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Circuit resistance (see 3.4). The circuit resistance shall be measured with approved resistance measuring equipment.

4.4.2 Insulation resistance (see 3.5). The insulation resistance shall be measured in accordance with table IV with approved resistance measuring equipment with a potential of 500 ± 50 volts direct current. The current shall be limited to 5 milliamperes. A minimum of 15 seconds shall be allowed before reading value after test connections have been made.

Table IV

Positive Pin	Negative
1	2

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The interval timer and power supply shall be packaged in accordance with dwg 8858967 and dwg 8808362 using the specified moisture barrier.

5.1.2 Level C. The interval timer and power supply shall be packaged in accordance with dwg 8858967 and dwg 8808362 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The interval timer and power supply shall be packed in accordance with dwg 8858967 and dwg 8808362.

5.3 Marking. Marking shall be as specified on dwgs 8858967 and 8808362.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

Custodian:
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Preparing activity:
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